

Spectral Energy Distribution Analyzer (SEDA) for Forward Modeling of Ultracool Objects



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Sherelyn Alejandro Merchan (CUNY)

Austin Rothermich (CUNY)

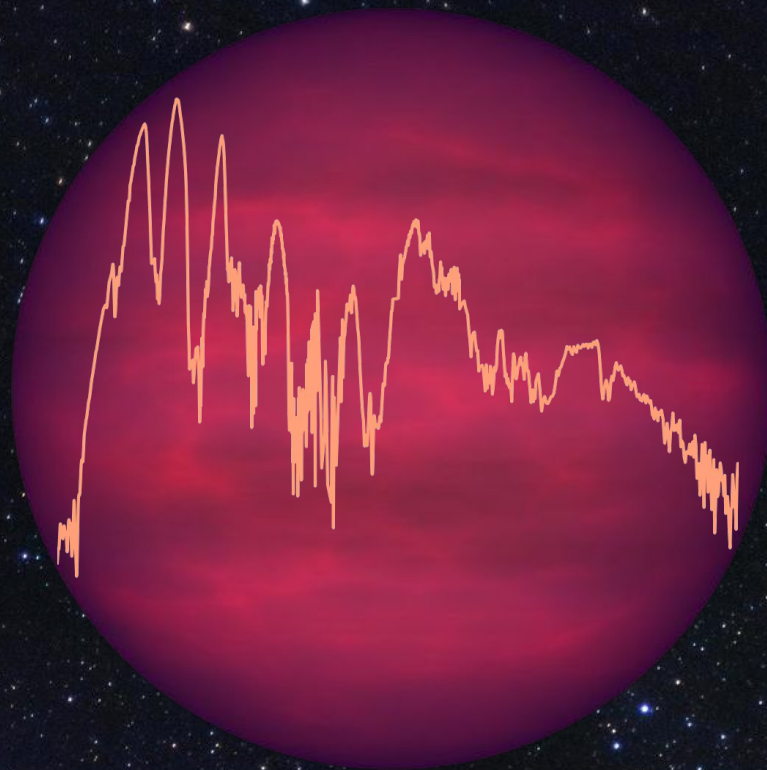
Daniella Bardalez Gagliuffi (Amherst College)

Niall Whiteford (AMNH)

245th AAS Meeting, National Harbor, Maryland. January 15, 2025.

SEDA Overview

Open-source Python package for forward modeling analysis of brown dwarfs, directly imaged exoplanets, and low-mass stars.

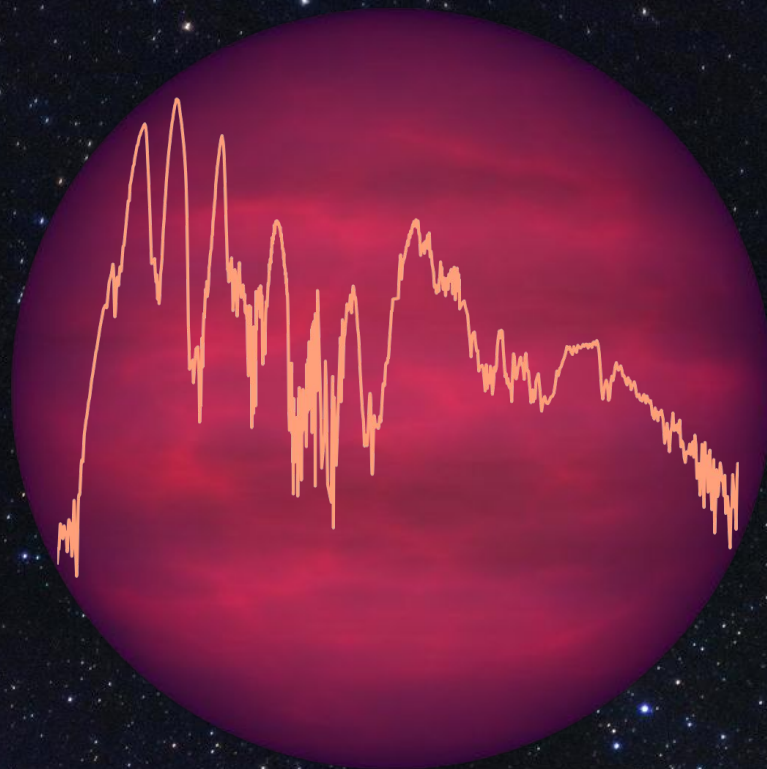


SEDA Overview

Open-source Python package for forward modeling analysis of brown dwarfs, directly imaged exoplanets, and low-mass stars.

Compares spectra and/or photometry to atmospheric models by using a Bayesian framework to sample posteriors.

Dynesty (Speagle 2020) for dynamic nested sampling.



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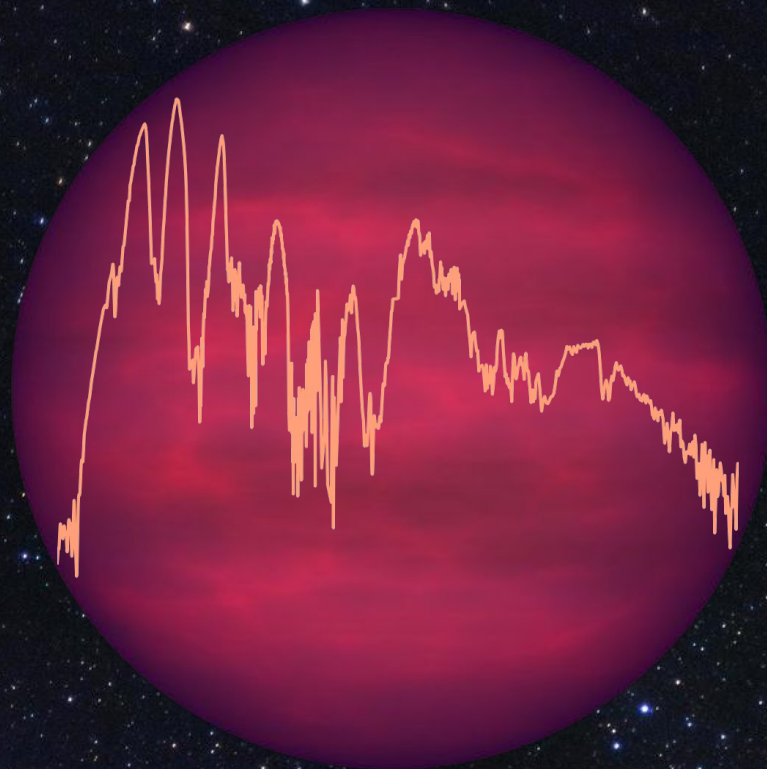
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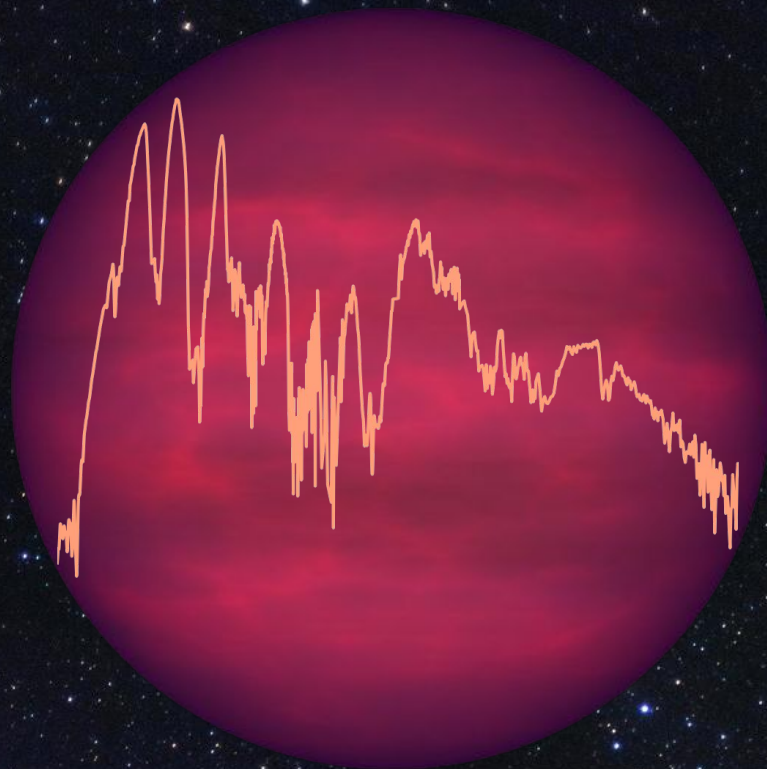
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Variety of functions to visualize results.



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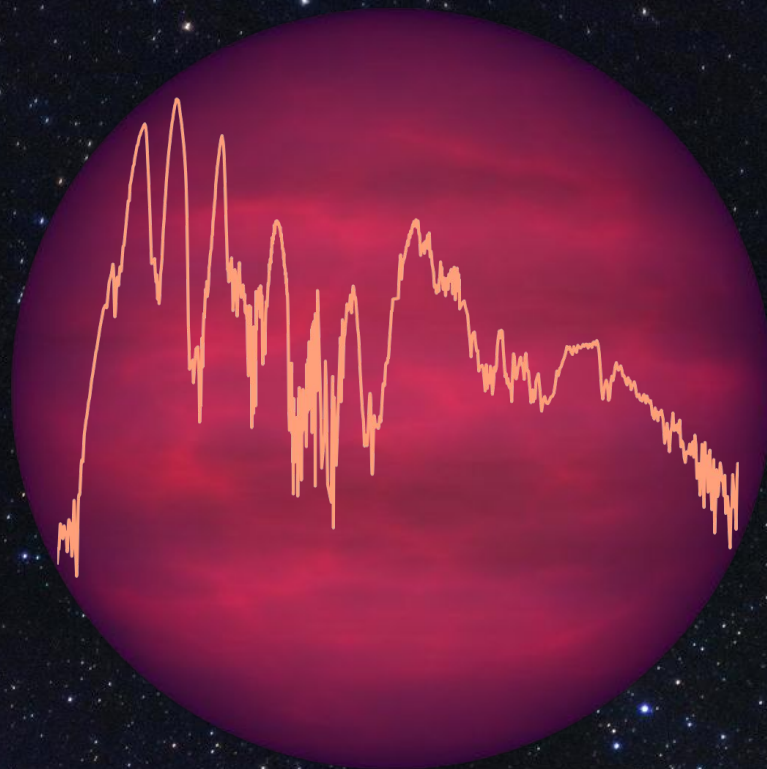
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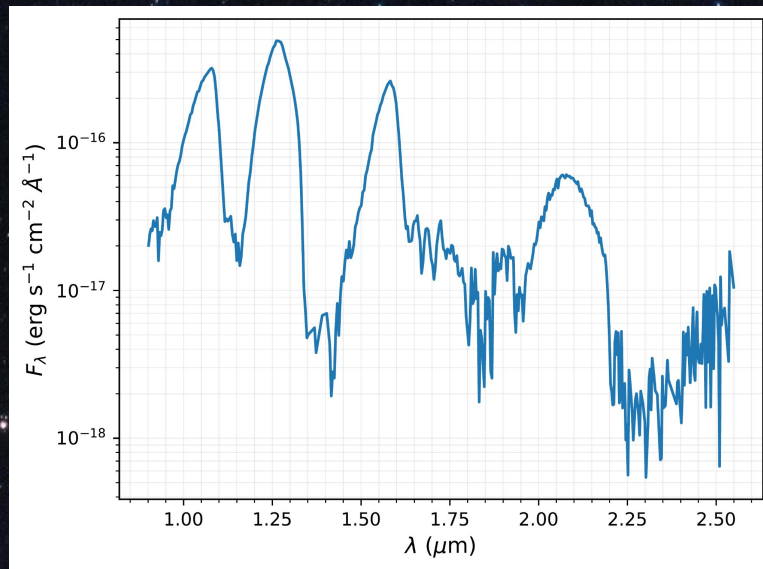
Several tools useful for the analysis of SEDs.



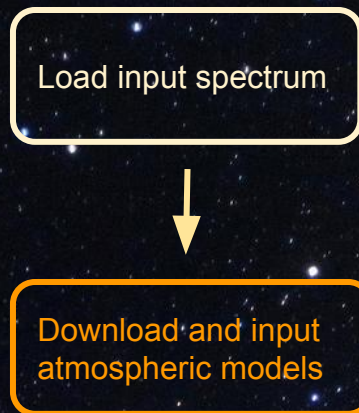
SEDA Workflow

Load input spectrum

IRTF/SpeX spectrum for the T8 (~750 K)
brown dwarf J0415 (Burgasser+2004):



SEDA Workflow



Available atmospheric models:

- Sonora Elf Owl models (Mukherjee et al. 2024)
- Sonora Diamondback (Morley et al. 2024)
- Lacy & Burrows (2023)
- Sonora Cholla (Karalidi et al. 2021)
- Sonora Bobcat (Marley et al. 2021)
- ATMO 2020 (Phillips et al. 2020)
- BT-Settl (Allard et al. 2012)
- Saumon & Marley (2008)

SEDA Workflow

Load input spectrum

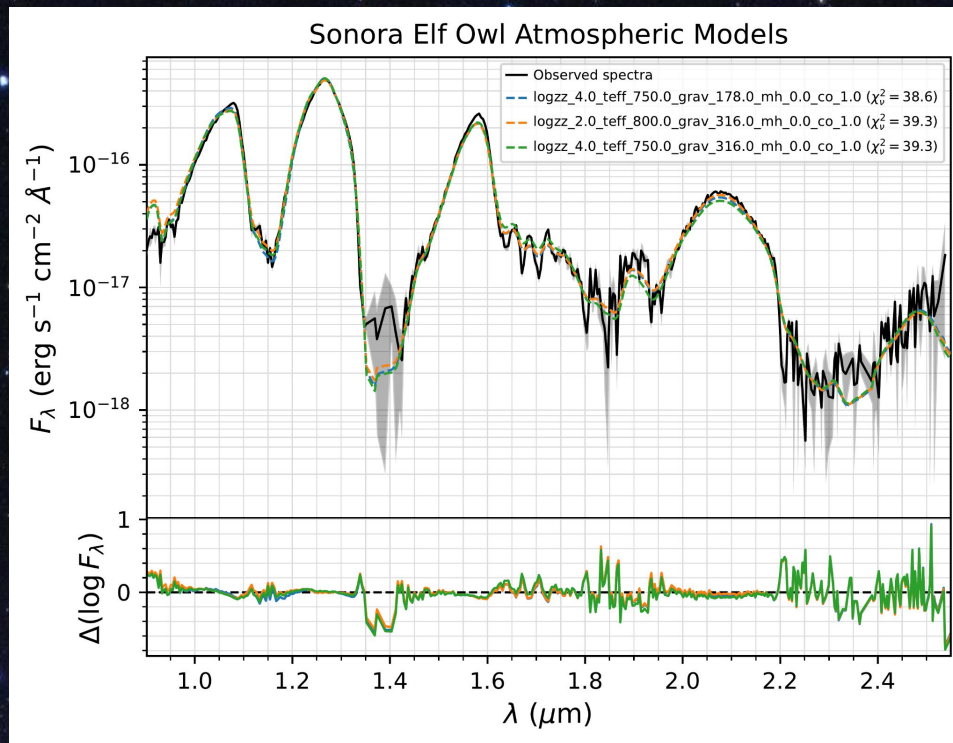


Download and input
atmospheric models

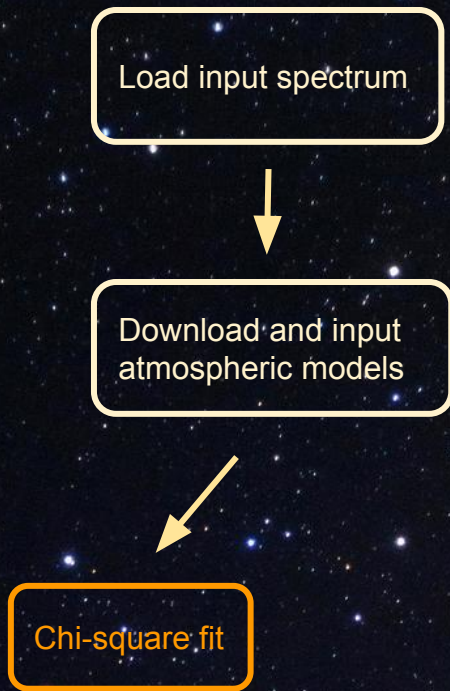


Chi-square fit

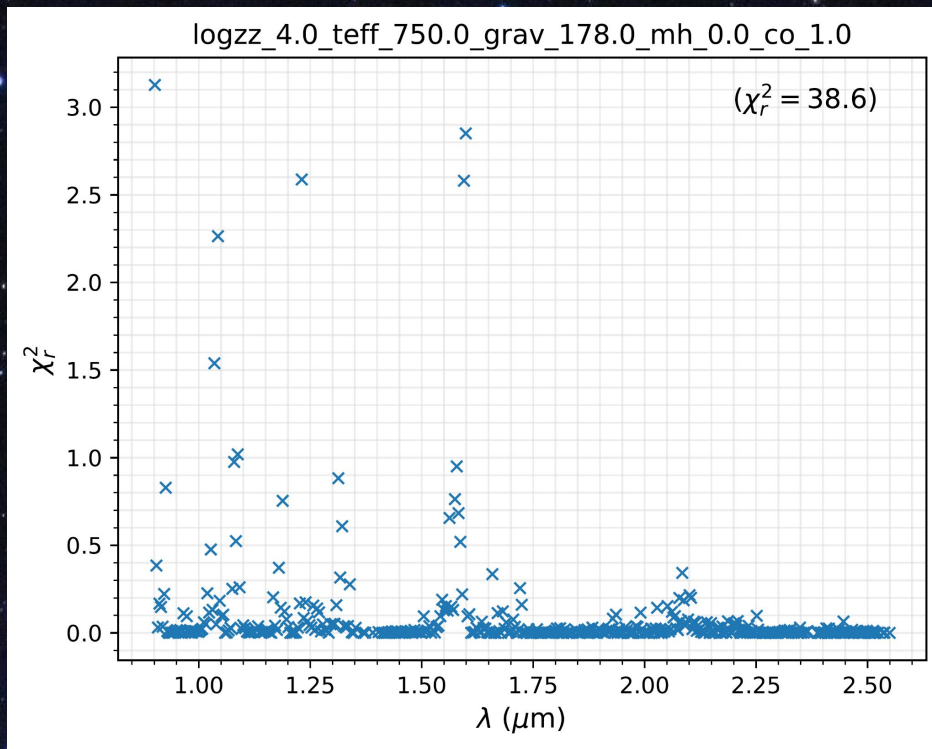
Best three model fits



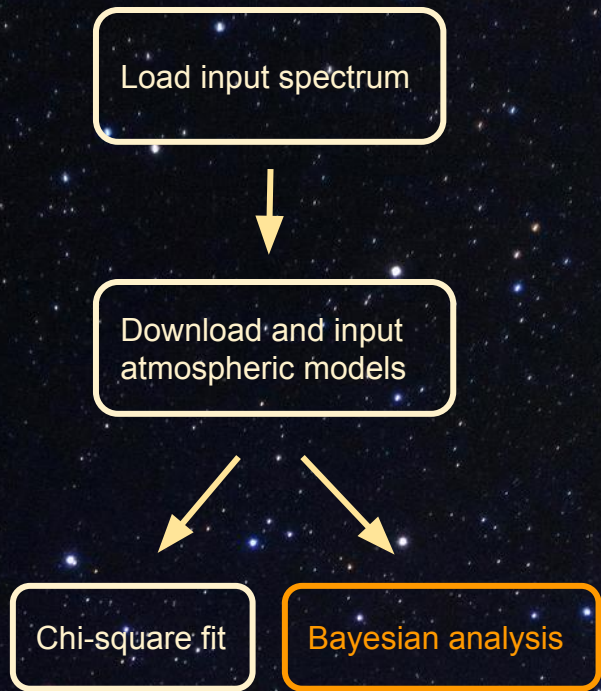
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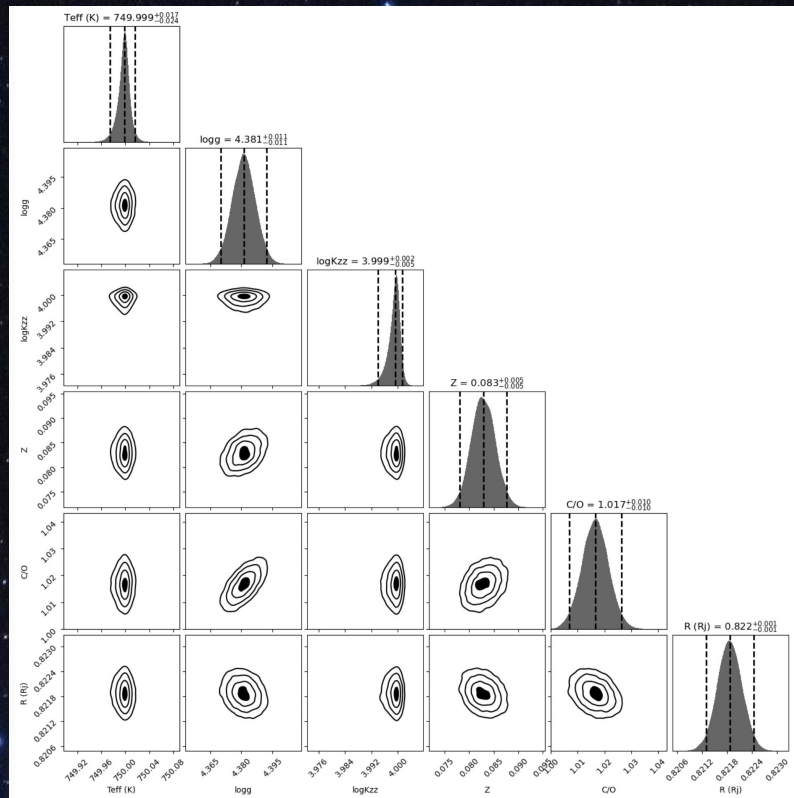
Reduced chi-square as a function of wavelength



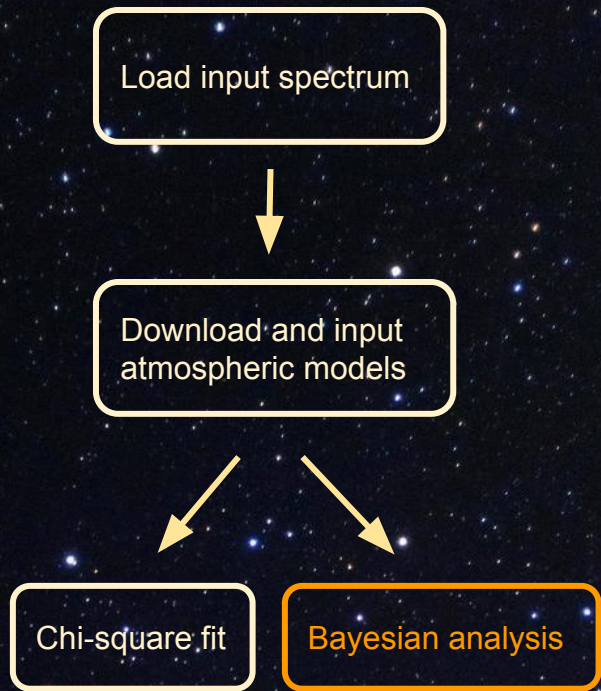
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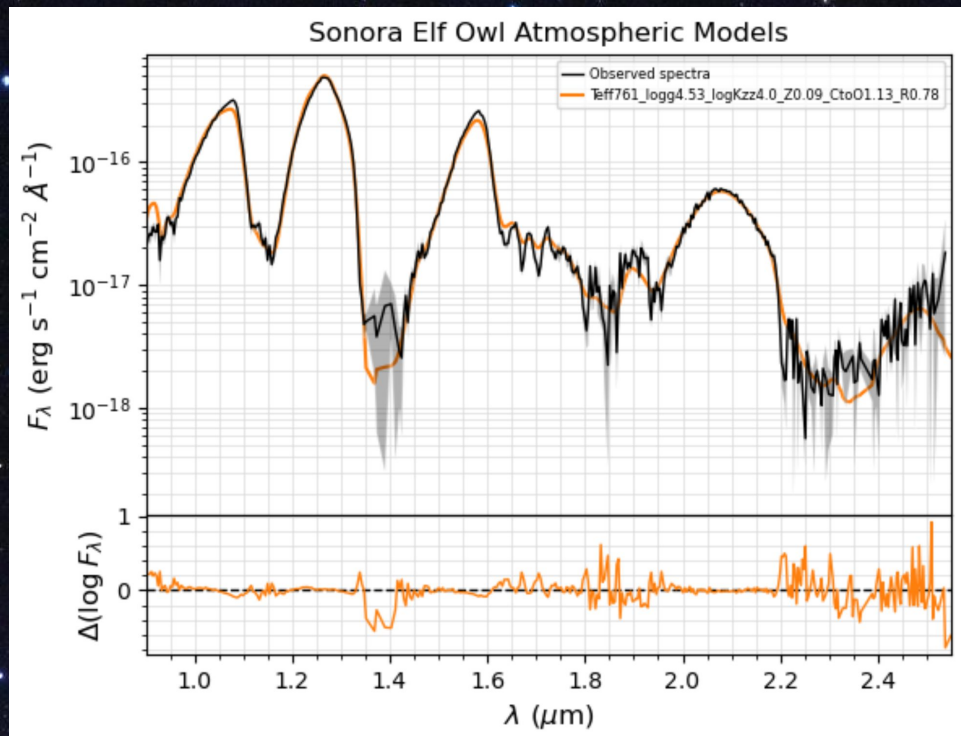
Corner plot with the posteriors



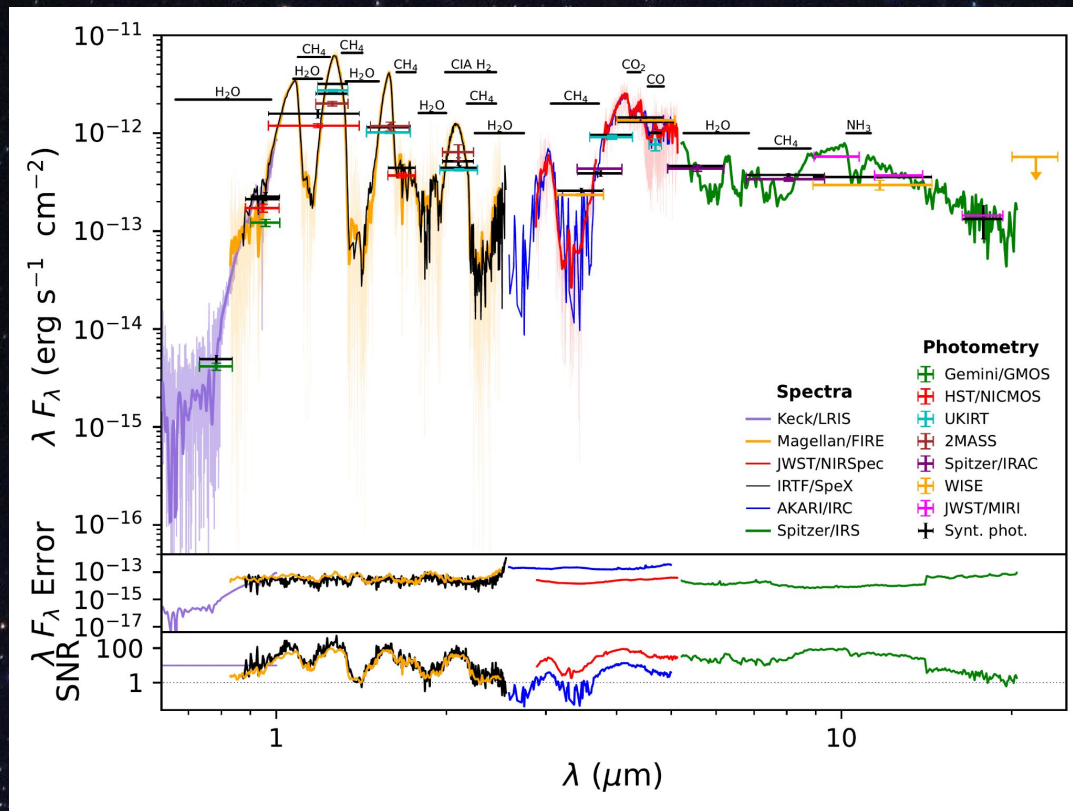
SEDA Workflow



Model with the median posteriors

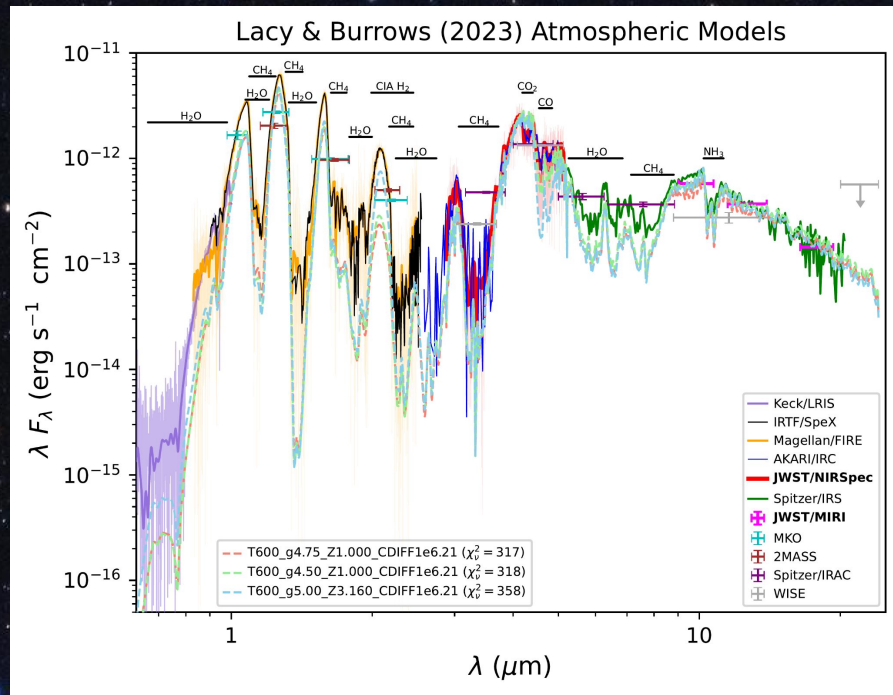
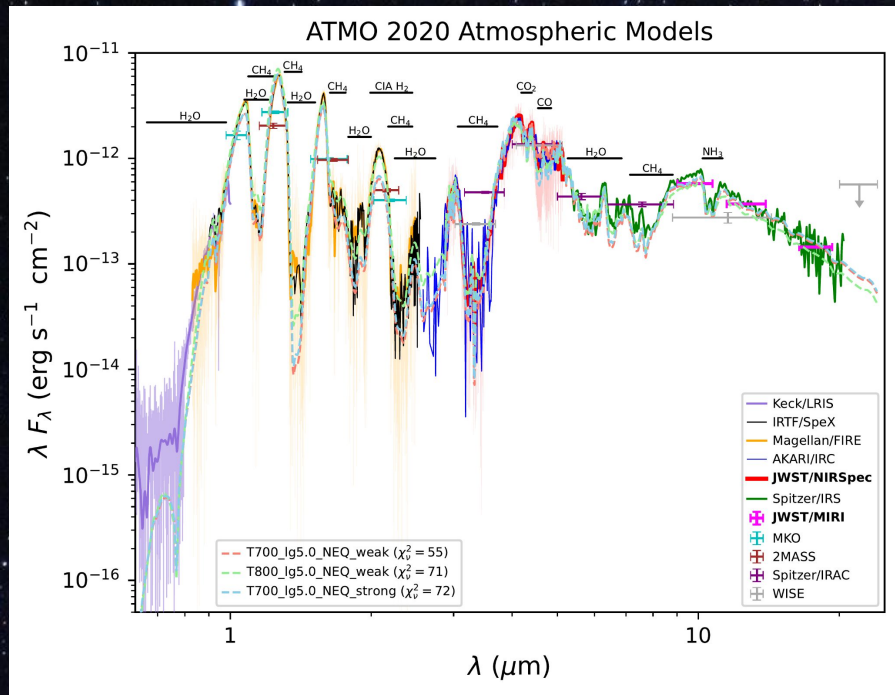


Modeling a Full SED



Alejandro Merchan et al. (2024, in prep.)

Modeling a Full SED



Suárez et al. (2025, in prep.)

Principal SEDA Tools

- Synthetic photometry for any SVO filters



Filter Profile Service

Carlos Rodrigo

A repository of Filter information for the VO.





VO Service

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Search text ~

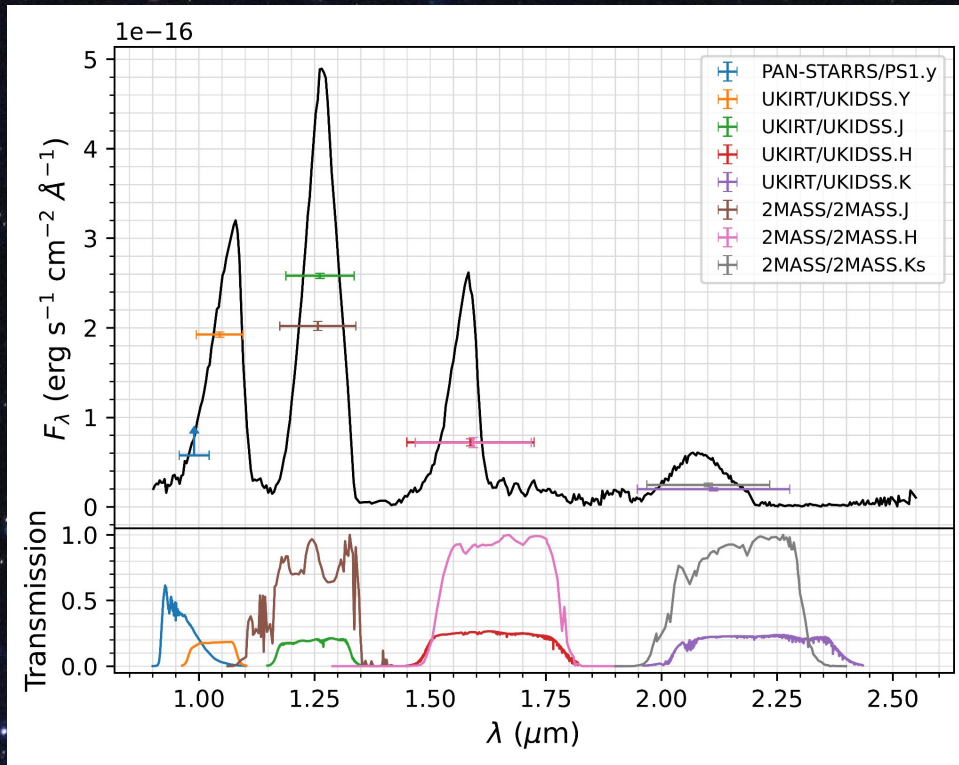
Search (?)

Astronomy (7905)

Planetary science (616)

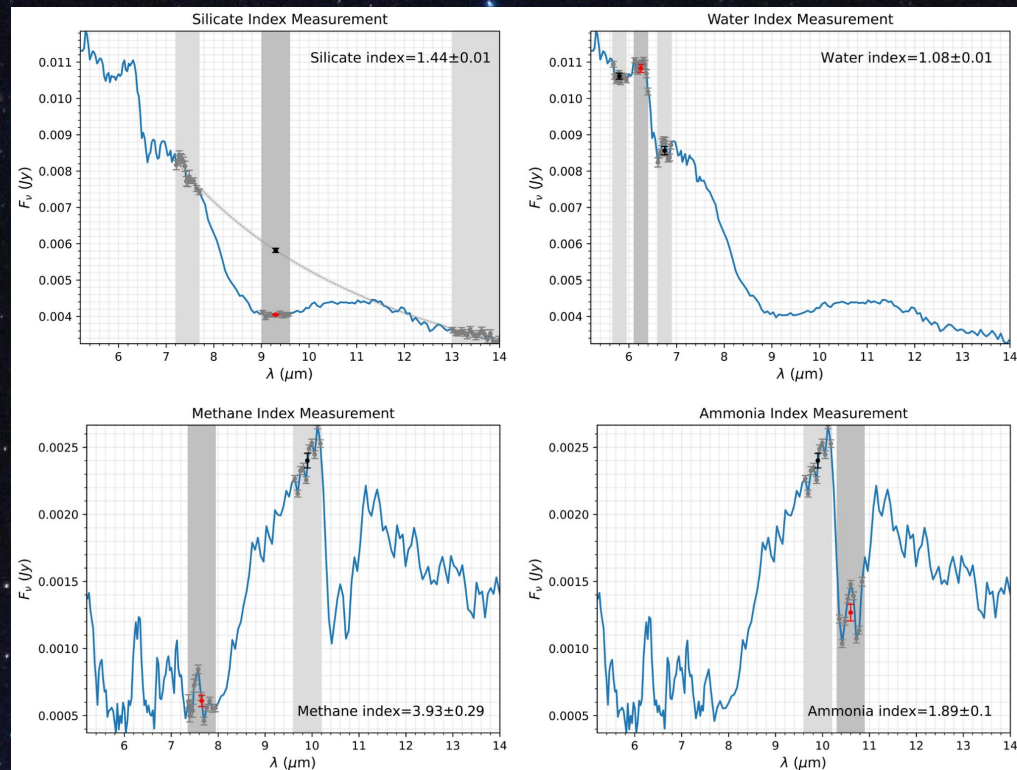
Earth Obs. (2452)

2MASS	AAG	ADIOS	AKARI	AKATSUKI	ALMA1	AFEX	APD	Aqua	ARCHEOPS	ARGO	Astronaut	Beijing1	Bepi-Colombo	BICEP	
BLAST	BOK	BOOMERANG	CAHA	Cameras	Cassini	CASTOR	CFHT	CHEOPS	Clementine	CNO-SAT	COBE	COMS	Contour	Corot	
COSMOSOMAS	CSST	CSTAR	CTIO	Dawn	DeepImpact	DENIS	DOT	DISCOVER	Envisat	ERBS	EROS	ERS	ESO	Euclid	
ExoMars	FengYun	Flock	FLWO	GAIA	GALEX	Galileo	GCOM-C	GCPD	Gemini	Generic	Geneva	GeoEye	Glottio	GOES	
GOTO	GTC	Hayabusa2	HCT	Herschel	Hinawari	Hipparcos	HST	IACB	IKONOS	ING	INSAT	Indisight	INT	Integral	
IRAM	IRAS	IRS	IRSP	IRTF	ISO	IUE	JCHT	JPS5	JWST	Keck	Kepler	KOMPSTAT	KPN	Landiat	
LacCumbres	LaSilla	LBT	LCO	LICK	Liverpool	LMT	LRO	LSST	LYRA	Mariner10	Mars2020	MAXIMA	MCD	MER	
Mercator	Messenger	Meteorat	METOP	MEX	Misc	MMT	MOA	MOH	MOI	MOIST	MRO	MSX	MT	NAOC	
NEAR	NewHorizons	NigeriaSat1	NIMBUS	NIRT	NOAA	NOAO	NOT	OAF	OAJ	DAN-SPM	OAO	Odyssey	OHF	OLIMPO	
OSIRIS-REX	OSN	OVID	P200	Pallomar	PAN-STARRS	Paranal	Paranal	Patfinder	PLANCK	PRINCE	PRINCE	Quickline	QUIET	QUOTE	
RapidEye	Roman	Rosetta	SALT	SAO	Scorpio	Sardinia	Selene	Sentinel	SEOSAT	SlyStar	SLOAN	SMART1	SOPH		
SOHO	SolarOrbiter	Special	SPECULOS	SPIDER	Spotter	SPOT	SPT	SSOT	Stardust	STELLA	Subaru	Swift	TAUVEX	TCS	
TD1	Terra	TESS	TIROS-N	TJO	TNG	TNO	TNT	Tophat	TRMM	TYCHO	UK-DMC	UKIRT	VATT	VenusExpress	
Viking	Voyager	WASP	WFIRST	WHT	WISE	WYN	WMAP	Worldview	XMM	Ziyuan					



Principal SEDA Tools

- Synthetic photometry
- Measure the depth of spectral features

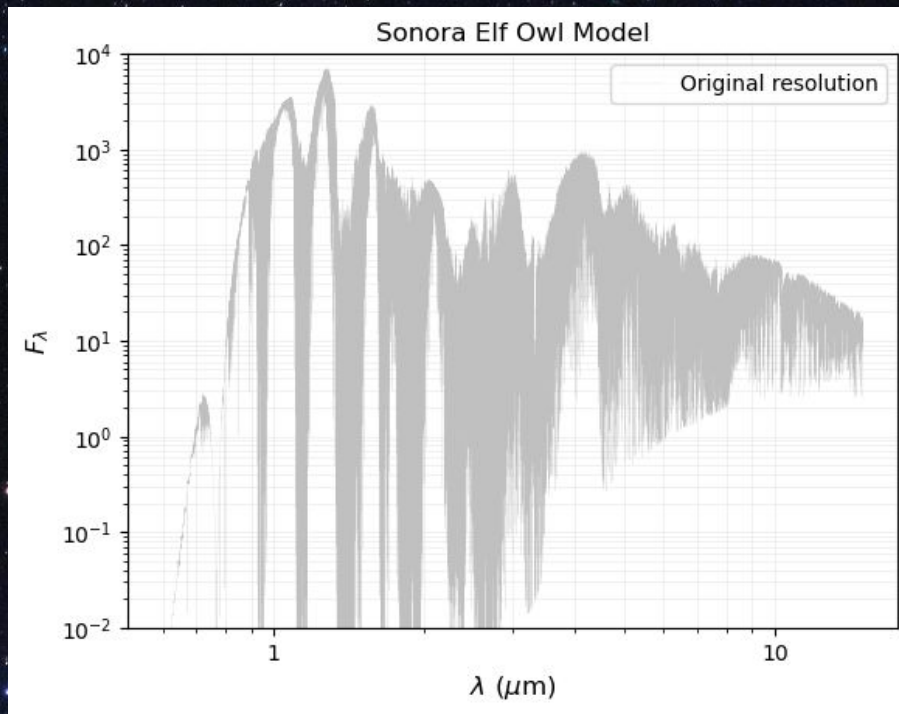


Principal SEDA Tools

- Synthetic photometry
- Measure the depth of spectral features
- Generate model spectra with parameters within the grid coverage

Sonora Elf Owl model:

$T_{\text{eff}}=789\text{ K}$, $\log g=4.38$, $\log K_{\text{zz}}=5.21$, $Z=-0.28$, $\text{C/O}=1.13$

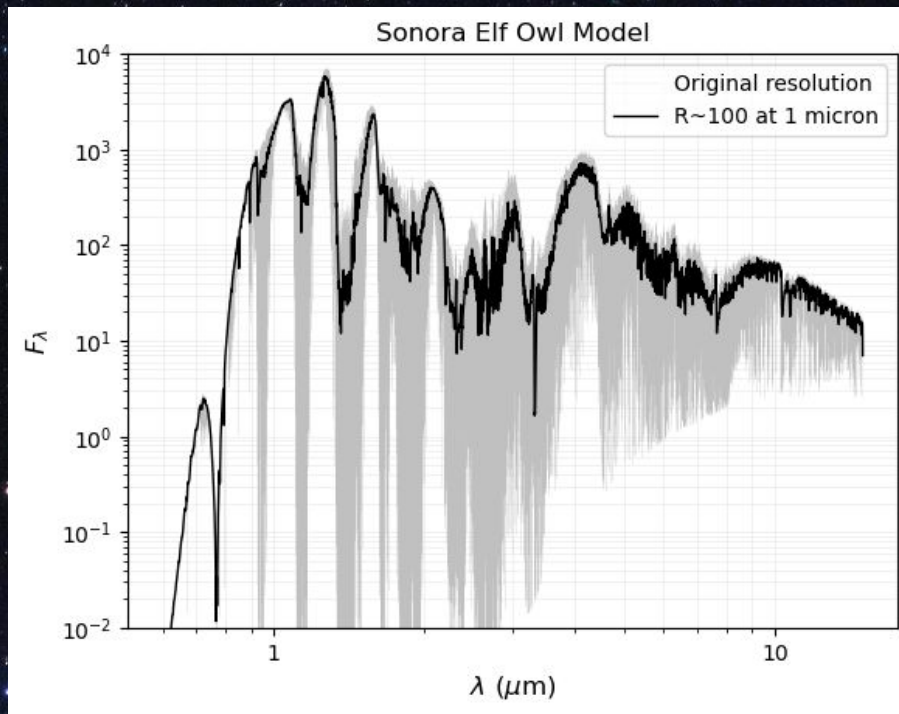


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- Convolve spectra to a desired resolution

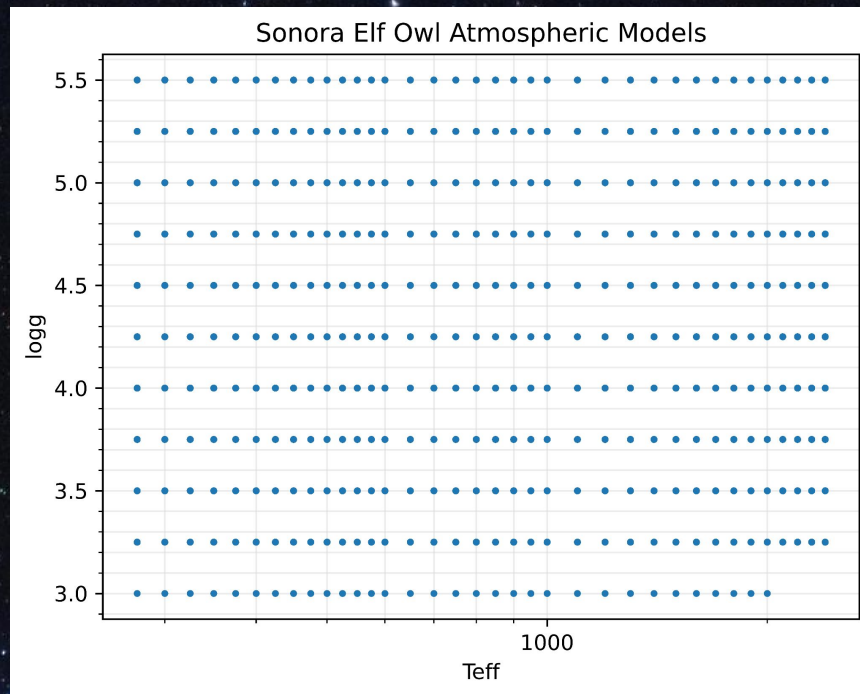
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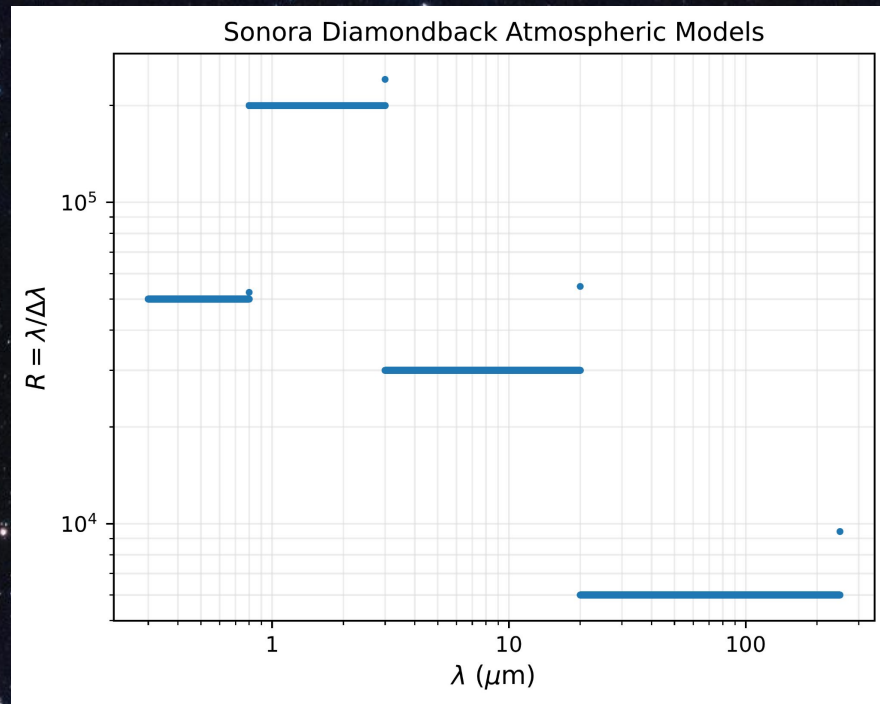
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- Measure the depth of spectral features
- Generate model spectra with parameters within the grid coverage
- Convolve spectra to a desired resolution
- Inspect model parameters coverage and resolution



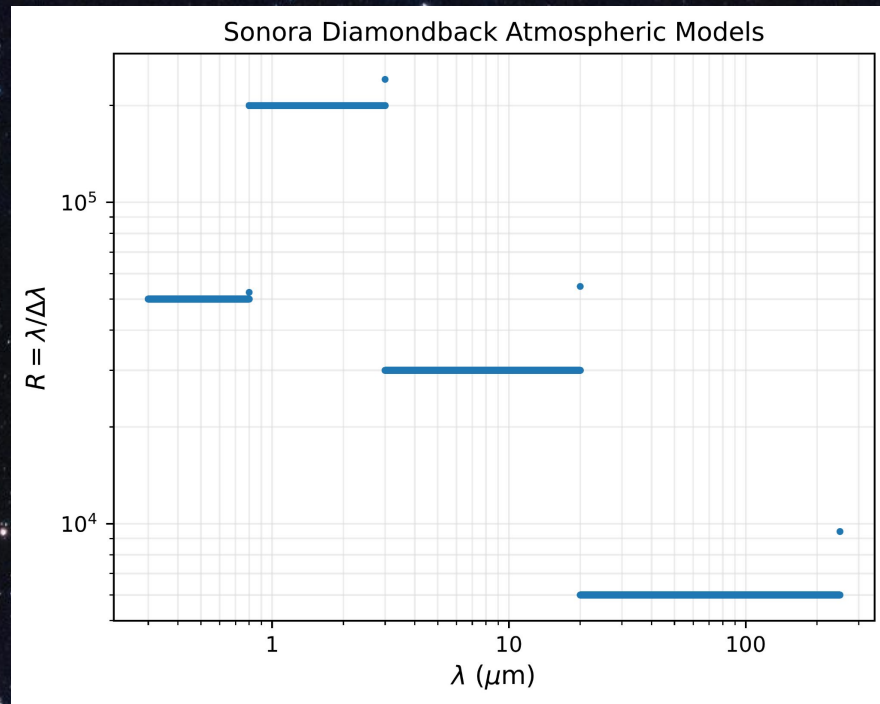
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- Measure the depth of spectral features
- Generate model spectra with parameters within the grid coverage
- Convolve spectra to a desired resolution
- Inspect model parameters coverage and resolution
- Convert fluxes into magnitudes and vice versa



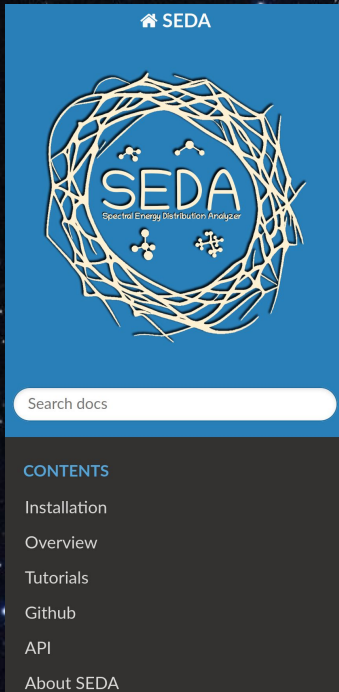
Takeaway

Check out the SEDA package to model and inspect SEDs of ultracool objects.



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Check out the content below for further information on how to install and use SEDA.

Contents

- Installation
 - Installation
 - Dependencies
- Overview
 - Principal Modules
 - Code Workflow
 - Main Tools
 - Available Atmospheric Models



• Tutorials

- Inspect available atmospheric models
- Compute synthetic photometry for any SVO filters
- Measure spectral indices from mid-infrared spectra
- Chi-square minimization for a single observed spectra
- Chi-square minimization for multiple observed spectra
- Nested sampling for a single observed spectrum
- Nested sampling for multiple spectra of the same object

• Github

• API

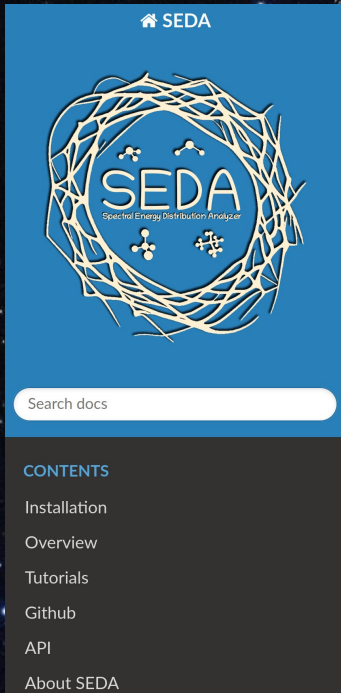
- Input Parameters
- Chi-square Fit
- Bayesian Sampling
- Utils
- Plots

• About SEDA

- Attribution
- Contributing
- Questions and feedback
- Logo
- FAQs

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Thanks for listening!